Winning Space Race with Data Science

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Executive Summary

This project analyzed SpaceX Falcon 9 launch data to identify the factors influencing first-stage landing success and to develop a predictive model for landing outcomes. The project followed a complete data science workflow, including data collection through APIs and web scraping, data wrangling, exploratory data analysis using SQL and visualization techniques, interactive mapping with Folium, dashboard development using Plotly Dash, and predictive modeling with machine learning.

Exploratory analysis showed that launch site, orbit type, payload mass, and booster version are associated with landing outcomes, while the yearly success rate demonstrated continuous improvements in Falcon 9 landing reliability. Interactive dashboards and maps enabled effective exploration of launch performance across different launch sites and mission characteristics.

Four classification models—Logistic Regression, Support Vector Machine (SVM), Decision Tree, and K-Nearest Neighbors (KNN)—were trained and evaluated. The Decision Tree model achieved the highest classification accuracy of 88.89%, making it the best-performing model for predicting Falcon 9 first-stage landing success.

Introduction

PROJECT BACKGROUND AND CONTEXT

- SpaceX has significantly reduced the cost of space launches by reusing the Falcon 9 first-stage booster.
- A Falcon 9 launch costs approximately $62 million, compared with more than $165 million for many competing launch providers.
- The ability to recover and reuse the first stage is a major reason for this cost advantage.
- Predicting whether the first stage will land successfully can help estimate the overall cost of a launch and support competitive decision-making.

PROBLEM TO BE SOLVED

- Analyze historical Falcon 9 launch data to identify the factors that influence first-stage landing success.
- Build machine learning classification models to predict whether the Falcon 9 first stage will land successfully.
- Compare different classification models and select the best-performing model based on evaluation metrics.

Section 1

Methodology

Methodology

Data Collection

Collected Falcon 9 launch data using the SpaceX REST API and web scraping techniques.

Data Wrangling

Cleaned, merged, and prepared the datasets by handling missing values and selecting relevant features.

Exploratory Data Analysis

Explored launch trends and relationships using SQL queries and data visualizations.

Interactive Visual Analytics

Built interactive maps with Folium and dashboards using Plotly Dash to analyze launch outcomes.

Predictive Analysis

Developed, tuned, and evaluated classification models to predict Falcon 9 first-stage landing success.

Model Evaluation

Compared model performance using evaluation metrics and selected the best-performing classifier.

Data Collection

Key Phrases -

- Collected Falcon 9 launch records from the - SpaceX dataset provided for the capstone project.

- Used Python and Pandas to load and inspect the launch data.

- Retrieved launch-related information from the project dataset and organized it into a structured DataFrame.

- Prepared the collected data for data wrangling, exploratory analysis, visualization, and predictive modeling.

Flow Chart –

Load Falcon 9 Launch Dataset → Import using Pandas → Inspect & Validate Dataset → Perform Data Wrangling → Prepare Dataset for EDA & Predictive Analysis

Data Collection – SpaceX API

Key Phrases -

- SpaceX launch records were used as the primary data source for this project.
- Python functions were implemented to retrieve launch-related information, including booster version, launch site, payload, and core details.
- The extracted information was organized into a structured dataset for further analysis.
- The prepared dataset was then used for data wrangling, exploratory analysis, and predictive modeling.

Flow Chart -

Retrieve Launch Records → Retrieve Launch Information - BoosterVersion, Payload, Launch Site, Core → Create Structured Dataset → Export dataset_part_1.csv →Data Wrangling

GitHub URL of the completed Data collection API notebook –

https://github.com/Volt2608/IBM-Data-Science-Capstone-SpaceX/blob/main/notebooks/01_Data_Collection_API.ipynb

Data Collection – Web Scraping

Key Phrases -

- Accessed the Falcon 9 launch records snapshot from Wikipedia.
- Retrieved the HTML page using the Python **Requests** library.
- Parsed the HTML content using **BeautifulSoup**.
- Extracted launch table headers and launch records.
- Converted the extracted records into a Pandas DataFrame.
- Exported the processed dataset for subsequent analysis.

Flow Chart –

Access Wikipedia Launch Records → Retrieve HTML (Python Requests) → Parse HTML (BeautifulSoup) → Extract Launch Records → Create Pandas DataFrame → Export **spacex_web_scraped.csv**

GitHub URL of the completed Web Scraping notebook -

https://github.com/Volt2608/IBM-Data-Science-Capstone-SpaceX/blob/main/notebooks/02_Web_Scraping.ipynb

Data Wrangling

Key Phrases -

- Loaded the prepared Falcon 9 launch dataset.
- Examined landing outcomes and mission features. Analyzed launch sites, orbit types, and mission outcomes.
- Explored launch sites, orbit types, and payload information.
- Created the binary landing **Class** variable.
- Prepared the dataset for predictive analysis.
- Exported **dataset_part_2.csv**.

Flow Chart –

Load dataset_part_1.csv → Inspect Data (Missing Values & Data Types) → Analyze Launch Sites, Orbit Types & Payload Information → Create Binary Landing Class (Target Variable) → Prepare Dataset for Predictive Analysis → Export dataset_part_2.csv

GitHub URL of the completed SpaceX Data Wrangling notebook –

https://github.com/Volt2608/IBM-Data-Science-Capstone-SpaceX/blob/main/notebooks/03_Data_Wrangling.ipynb

EDA with Data Visualization

Key Phrases -

- **Scatter Plots:** Examined the relationships between **Flight Number, Payload Mass, Launch Site,** and **Orbit** to identify patterns associated with first-stage landing success.
- **Bar Chart:** Compared **landing success rates across different orbit types** to evaluate which mission orbits achieved higher landing success.
- **Line Chart:** Visualized the **yearly trend of Falcon 9 landing success** to observe improvements in landing performance over time.
- The visualizations were used to identify **patterns, correlations, and factors influencing first-stage landing success**.
- These visualizations helped identify **trends, correlations, and key factors** influencing Falcon 9 first-stage landing success.

GitHub URL of the completed EDA with Data Visualization notebook –

https://github.com/Volt2608/IBM-Data-Science-Capstone-SpaceX/blob/main/notebooks/04_EDA_Visualization.ipynb

EDA with SQL

Summary of SQL Queries -

- Retrieved **distinct launch sites** and filtered launch records using launch site, landing outcome, payload mass, and date conditions.
- Calculated **aggregate statistics** such as total, average, and maximum payload mass to evaluate mission performance.
- Identified **mission milestones**, including the first successful ground-pad landing and booster versions with the highest payload capacity.
- Used **GROUP BY** and **ORDER BY** clauses to summarize and rank mission outcomes across launch sites and booster versions.
- Applied **subqueries** to retrieve missions with the maximum payload and compare mission performance.
- Used **string functions** (e.g., SUBSTR) to analyze landing outcomes over a specified time period.

GitHub URL of the completed EDA with SQL notebook –

https://github.com/Volt2608/IBM-Data-Science-Capstone-SpaceX/blob/main/notebooks/05_EDA_SQL.ipynb

Build an Interactive Map with Folium

Summary of Map Objects –

- **Markers:** Added to identify each **Falcon 9 launch site** and display launch site names using interactive labels.
- **Circles:** Used to highlight launch site locations and improve map visibility.
- **Marker Clusters:** Grouped launch records at the same location to reduce overlap and improve readability.
- **Color-coded Markers:** Used **green** markers for successful landings and **red** markers for unsuccessful landings to visualize mission outcomes.
- **Polylines:** Drew lines between launch sites and nearby coastlines to analyze their geographic proximity.

Purpose -

These interactive map elements were used to **visualize launch site locations, compare mission outcomes geographically, reduce marker overlap, and analyze the spatial characteristics of Falcon 9 launch operations.**

GitHub URL of the completed Folium notebook –

https://github.com/Volt2608/IBM-Data-Science-Capstone-SpaceX/blob/main/notebooks/06_Folium_Map.ipynb

Build a Dashboard with Plotly Dash

Summary of Dashboard Components –

- **Launch Site Dropdown:** Enabled users to select **all launch sites** or a **specific launch site** for interactive analysis.
- **Pie Chart:** Displayed the **total successful launches by site** or the **success versus failure distribution** for a selected launch site.
- **Payload Range Slider:** Allowed users to filter launches by **payload mass** and dynamically update the dashboard.
- **Scatter Plot:** Visualized the relationship between **payload mass and landing success**, with booster version categories distinguished by color.
- **Interactive Callbacks:** Connected user selections to the charts, enabling real-time updates and interactive data exploration.

Purpose –

The interactive dashboard was designed to **explore launch performance, compare success rates across launch sites, analyze the effect of payload mass on landing success, and provide an interactive interface for data exploration.**

GitHub URL of the completed SpaceX Ploty Dash notebook –

https://github.com/Volt2608/IBM-Data-Science-Capstone-SpaceX/blob/main/dashboard/spacex-dash-app.py

Predictive Analysis (Classification)

Summary –

Built and evaluated multiple machine learning classification models to predict Falcon 9 first-stage landing success. The workflow included data preprocessing, feature scaling, train-test splitting, hyperparameter tuning using GridSearchCV, and model evaluation to identify the best-performing classifier.

Key Phrases –

- Loaded and prepared the dataset for classification.
- Split the data into **training** and **testing** sets.
- Standardized features using **StandardScaler**.
- Trained **Logistic Regression, SVM, Decision Tree,** and **KNN** models.
- Optimized model parameters using **GridSearchCV (10-fold cross-validation)**.
- Evaluated model performance using **confusion matrices** and selected the best-performing model.

GitHub URL of the completed SpaceX Predictive Analysis notebook –

https://github.com/Volt2608/IBM-Data-Science-Capstone-SpaceX/blob/main/notebooks/07_Predictive_Analysis.ipynb

Results

Exploratory Data Analysis Results -

- Identified relationships between **payload mass, orbit type, launch site, and landing success**.
- Observed an overall improvement in **Falcon 9 landing success over time**.
- SQL analysis provided additional insights into **launch performance and mission characteristics**.

Interactive Analytics Demo -

- Demonstrated **interactive launch site analysis** using Folium maps.
- Developed a **Plotly Dash dashboard** for interactive exploration of launch success and payload trends.

Predictive Analysis Results –

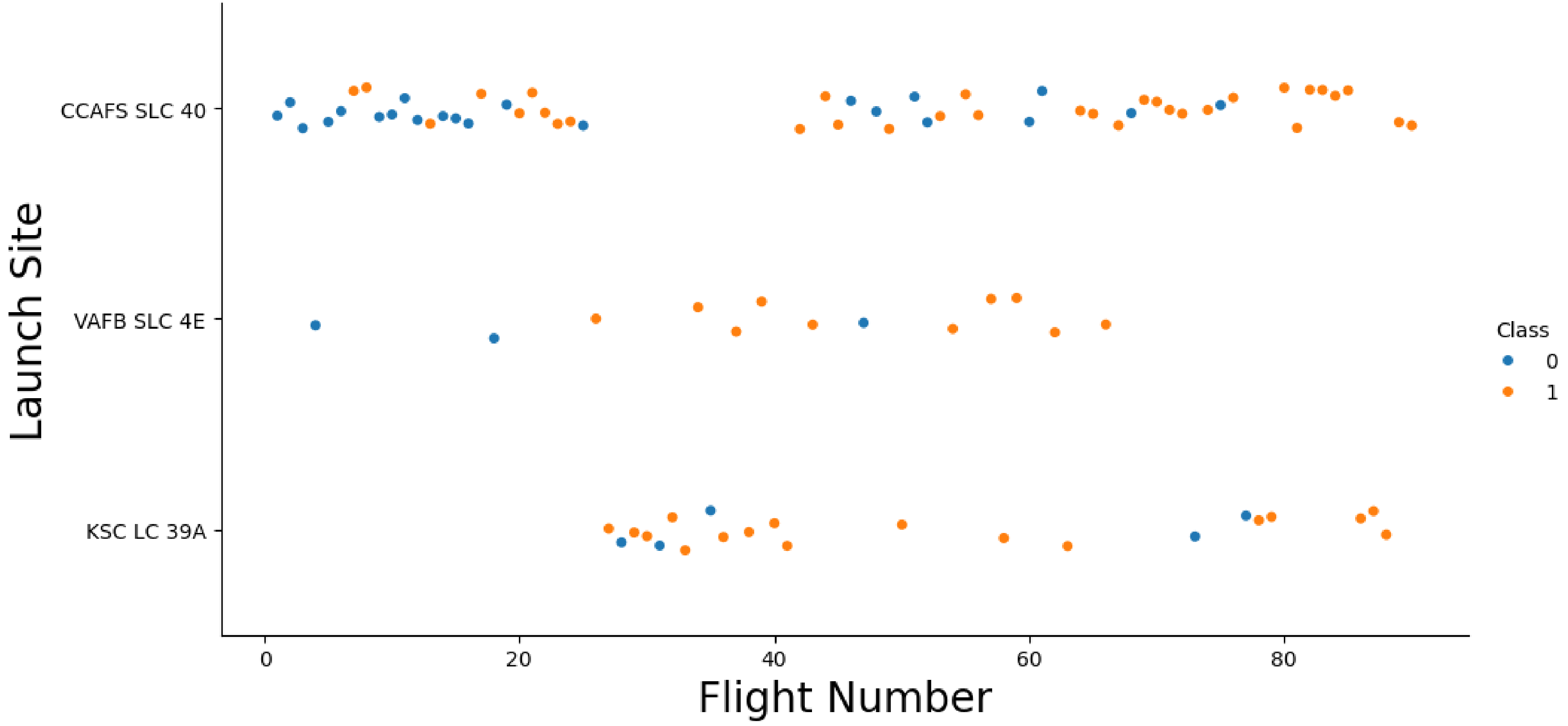
- Built and evaluated **Logistic Regression, SVM, Decision Tree**, and **KNN** classification models.
- Optimized model performance using **GridSearchCV**.
- Selected the **best-performing model** based on evaluation metrics.

Section 2

Insights drawn from EDA

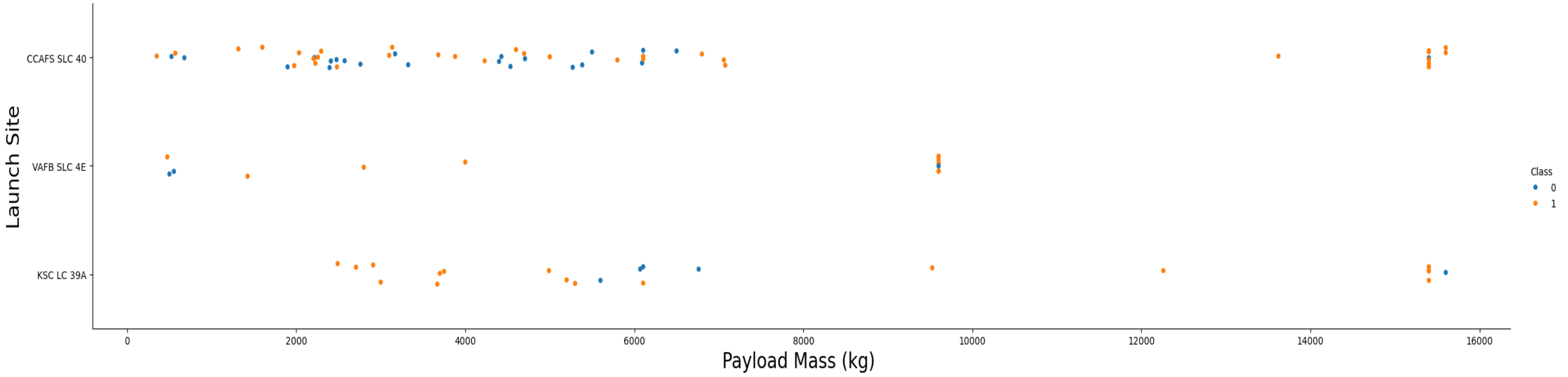
Scatter Plot And Explanation for Flight Number vs. Launch Site Analysis

- The scatter plot compares **flight number** across different **launch sites** while distinguishing successful and unsuccessful landings.
- **CCAFS SLC 40** hosted the largest number of Falcon 9 launches throughout the dataset
- Successful and unsuccessful missions are observed at all launch sites, with later flight numbers generally showing more successful landings.



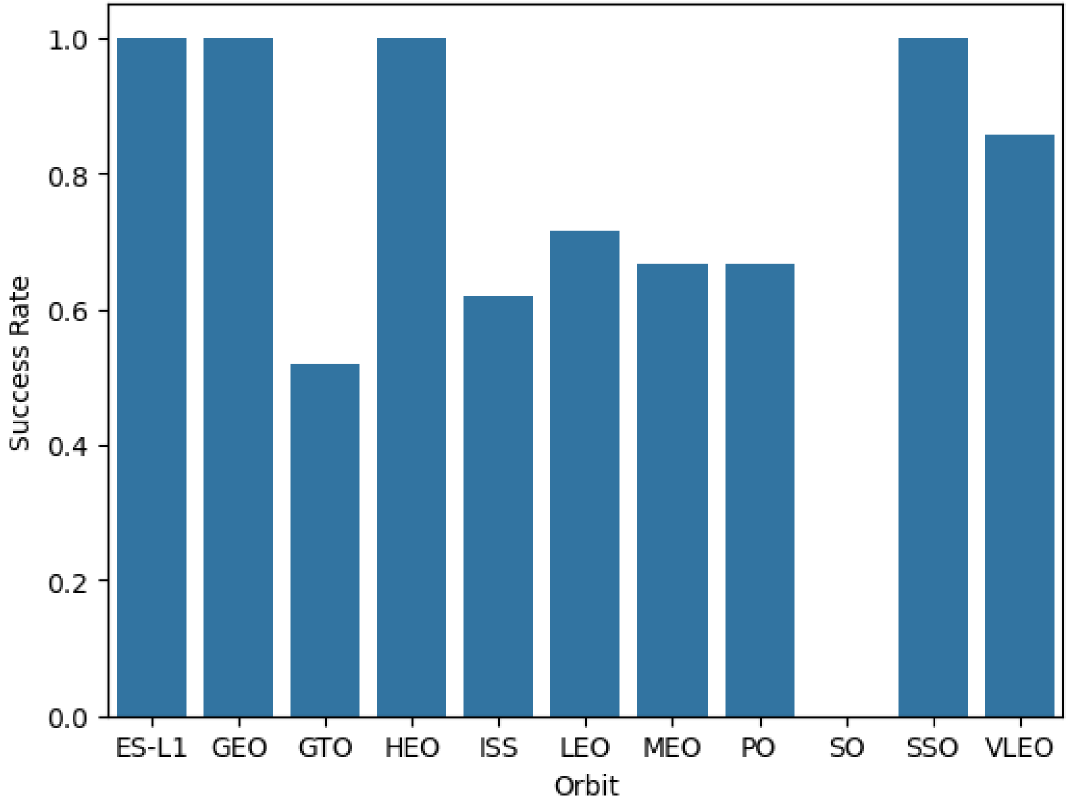
Scatter Plot And Explanation for Payload Mass vs. Launch Site Analysis

The scatter plot was used to analyze the relationship between payload mass and launch site while distinguishing successful and unsuccessful first-stage landings. It shows that payload capacity varies across launch sites. **VAFB SLC-4E** did not launch heavy payloads (greater than **10,000 kg**), whereas **CCAFS SLC-40** and **KSC LC-39A** supported a wider range of payload masses, including heavier missions. The visualization also indicates that successful landings occurred across multiple payload ranges, suggesting that launch site and mission characteristics both influence landing outcomes.



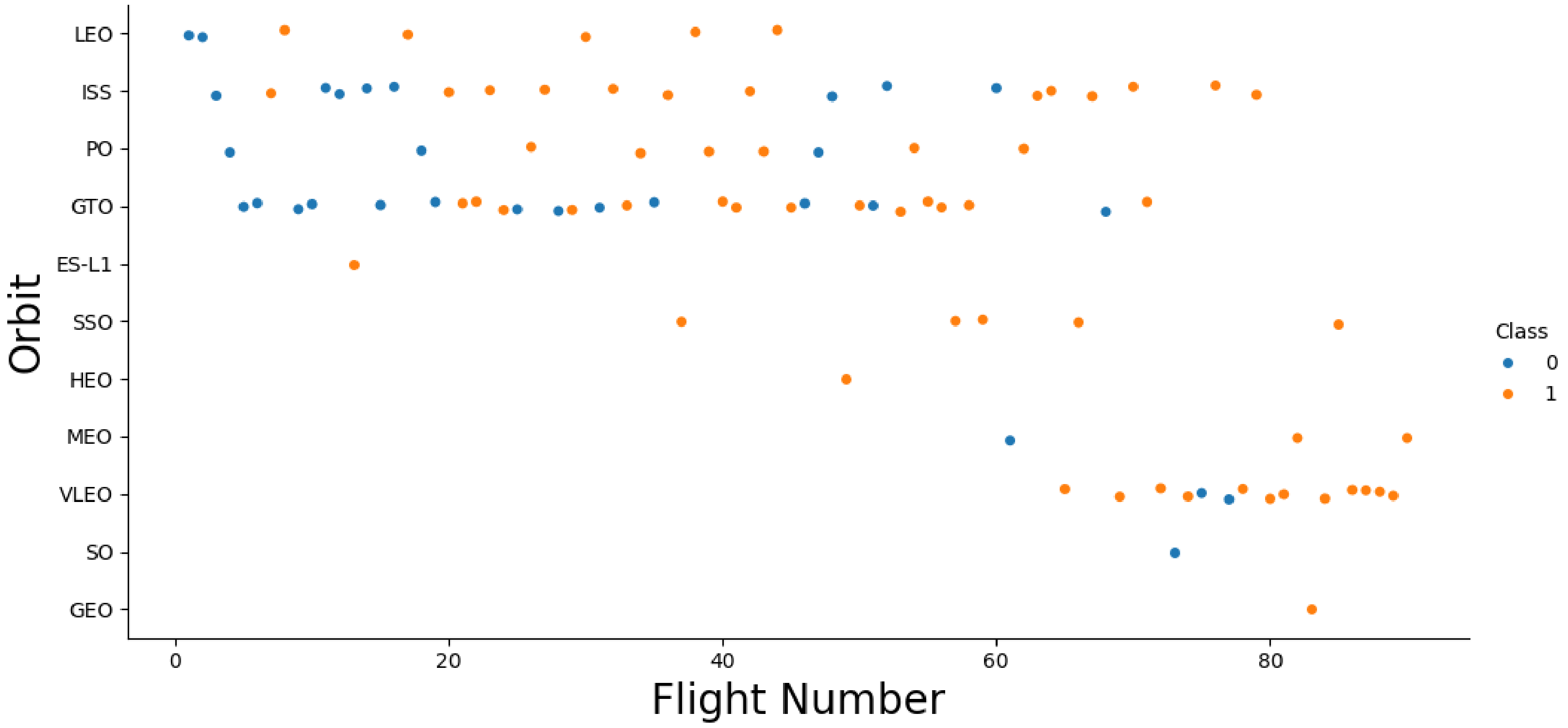
Bar Chart And Explanation for Success Rate vs. Orbit Type Analysis

The bar chart compares the first-stage landing success rate across different orbit types. It shows that **ES-L1, GEO, HEO, and SSO** missions achieved a **100% landing success rate**, while **GTO** missions had the **lowest success rate**. **LEO** and **ISS** missions demonstrated relatively high success rates because they had more launches and successful recoveries. This visualization indicates that mission orbit influences the probability of a successful first-stage landing.



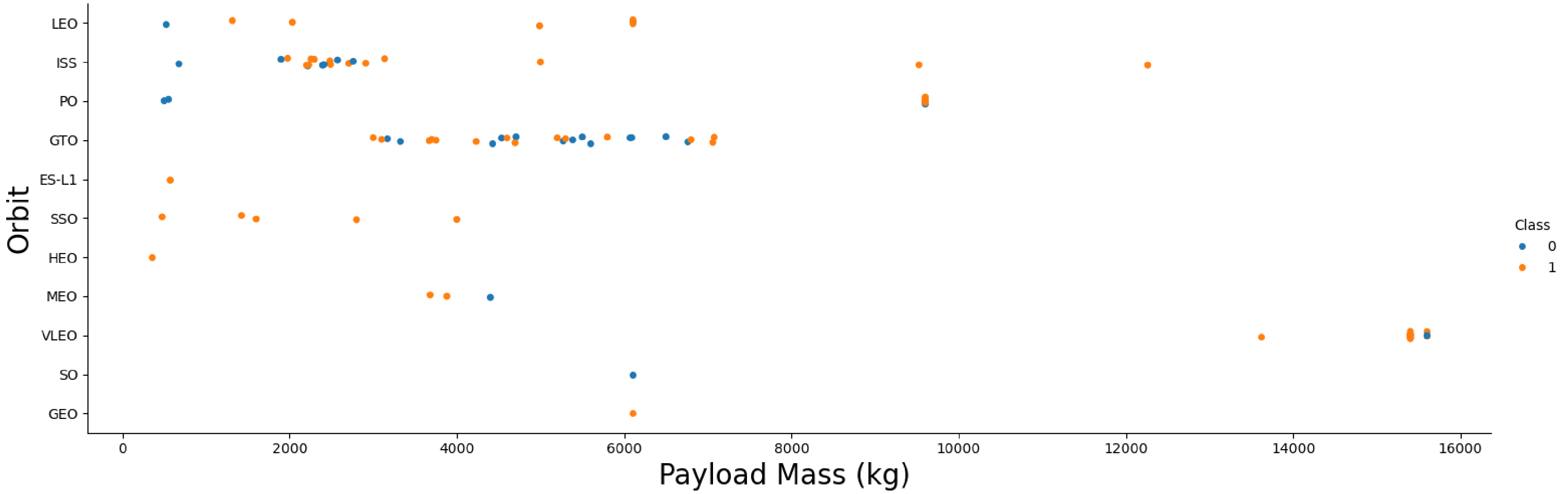
Scatter Plot And Explanation for Flight Number vs. Orbit Type Analysis

The scatter plot illustrates the relationship between Flight Number and Orbit Type, with point colors representing the landing outcome (successful or unsuccessful). As the flight number increases, SpaceX launches are distributed across multiple orbit types, with successful landings becoming more frequent in later missions. The visualization indicates an overall improvement in landing success across different orbit categories over successive Falcon 9 missions.



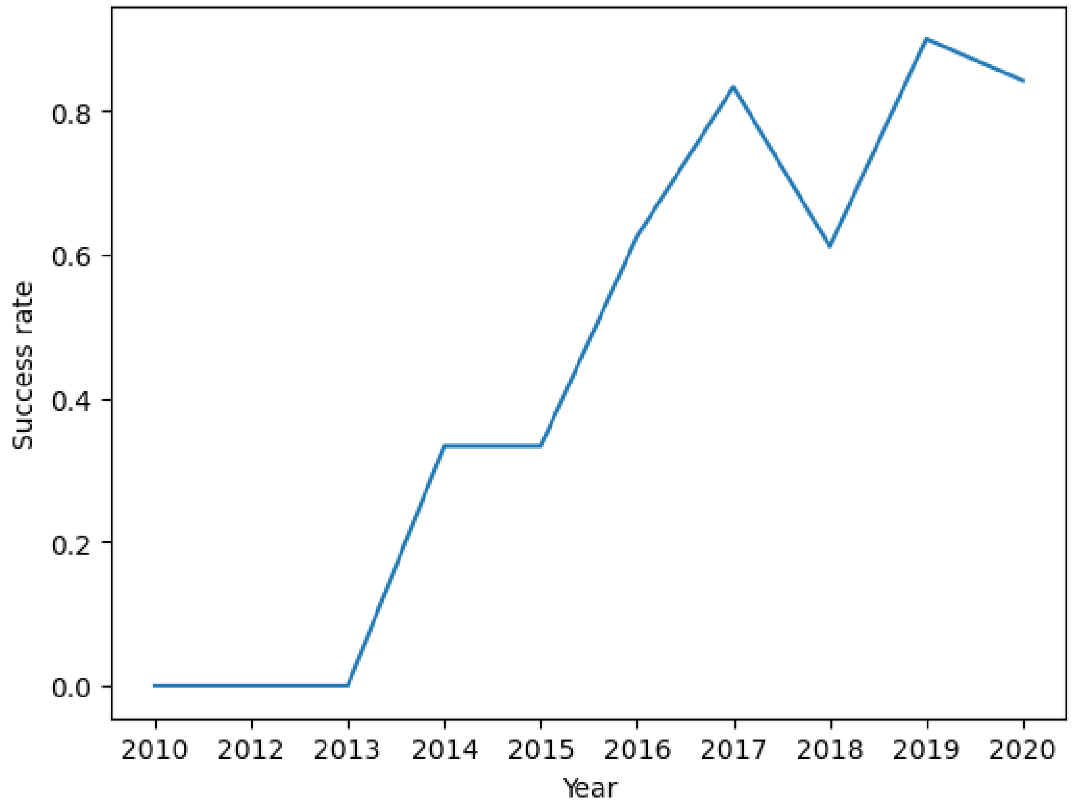
Scatter Plot And Explanation for Payload vs. Orbit Type Analysis

The scatter plot illustrates the relationship between Payload Mass and Orbit Type, with point colors representing successful and unsuccessful first-stage landings. Payload masses vary across different orbit types, while successful landings are observed over a broad range of payload values. The visualization indicates that landing success occurs across multiple orbit categories and is not determined by payload mass alone.



Line Chart And Explanation for Launch Success Yearly Trend

The line chart illustrates the yearly average success rate of Falcon 9 first-stage landings. The success rate shows an overall upward trend over successive years, indicating continuous improvements in landing performance. Although some yearly fluctuations are observed, the visualization demonstrates that SpaceX achieved increasingly reliable first-stage recoveries as more missions were completed.



The names of the unique launch sites

Unique launch sites -

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Presenting query result with a short explanation here –

The SQL query used the DISTINCT keyword to retrieve the unique launch sites from the SpaceX dataset. The results show four launch site names used across Falcon 9 missions, providing the basis for comparing launch activity and mission outcomes at different locations.

Launch Site Names Begin with 'CCA'

Presenting query result with a short explanation here –

The SQL query used the LIKE 'CCA%' operator to retrieve the first five launch records from launch sites whose names begin with "CCA". The results show that all five records correspond to launches from CCAFS LC-40, including early Falcon 9 demonstration and Commercial Resupply Services (CRS) missions, illustrating the use of pattern matching for filtering launch site records.

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

Presenting query result with a short explanation here –

The SQL query uses the sum() aggregate function to calculate the total payload mass carried for NASA (CRS) missions. The result shows that Falcon 9 boosters transported a cumulative payload of **45,596 kg** for NASA Commercial Resupply Services (CRS) missions, demonstrating the significant contribution of Falcon 9 to ISS cargo transportation.

Total_Payload
45596

Average Payload Mass by F9 v1.1

Presenting query result with a short explanation here –

The SQL query uses the AVG() aggregate function to calculate the average payload mass carried by the Falcon 9 v1.1 booster. The result shows that the average payload mass for missions using this booster version was **2,928.4 kg**, providing a summary measure of its payload-carrying performance across all recorded launches.

Average_Payload
2928.4

First Successful Ground Landing Date

Presenting query result with a short explanation here –

The SQL query uses the MIN() function to identify the earliest date on which a Falcon 9 first stage achieved a successful landing on a ground pad. The result shows that the first successful ground pad landing occurred on **22 December 2015**, marking a major milestone in SpaceX's reusable rocket program.

First_Success
2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

Presenting query result with a short explanation here –

The SQL query filters missions that achieved a successful drone ship landing with a payload mass between **4,000 kg** and **6,000 kg**. The results identify four Falcon 9 Full Thrust (F9 FT) booster versions—**B1022, B1026, B1021.2, and B1031.2**—that satisfied both the landing outcome and payload mass criteria.

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

Presenting query result with a short explanation here –

The SQL query groups mission outcomes and counts the number of occurrences for each category using the COUNT() aggregate function. The results show that out of 101 recorded missions, **99** were successful, **1** resulted in an in-flight failure, and **1** mission had a successful launch with an unclear payload status, indicating a very high overall mission success rate.

TRIM("Mission_Outcome")	Total
Failure (in flight)	1
Success	99
Success (payload status unclear)	1

Boosters Carried Maximum Payload

Presenting query result with a short explanation here –

The SQL query identifies all booster versions that carried the maximum payload mass by comparing each mission's payload mass with the maximum payload value in the dataset. The results show 12 Falcon 9 Block 5 (F9 B5) booster versions that achieved the highest recorded payload mass, demonstrating the capability of the Block 5 variant to support SpaceX's heaviest payload missions.

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

2015 Launch Records

Presenting query result with a short explanation here –

The SQL query filters Falcon 9 missions launched in **2015** with a **failed drone ship landing**. The results show two such missions: one in **January** using booster **F9 v1.1 B1012** and another in **April** using booster **F9 v1.1 B1015**, both launched from **CCAFS LC-40**. This query demonstrates how multiple filtering conditions can be combined to analyze mission outcomes for a specific year and landing type.

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Presenting query result with a short explanation here –

The SQL query counts and ranks all landing outcomes for Falcon 9 missions between **4 June 2010** and **20 March 2017**. The results show that **"No attempt"** was the most frequent landing outcome with **10** occurrences, followed by **5 successful** and **5 failed** drone ship landings. The ranking provides an overview of landing performance during the early years of Falcon 9 missions and highlights the gradual transition toward successful booster recovery.

Landing_Outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

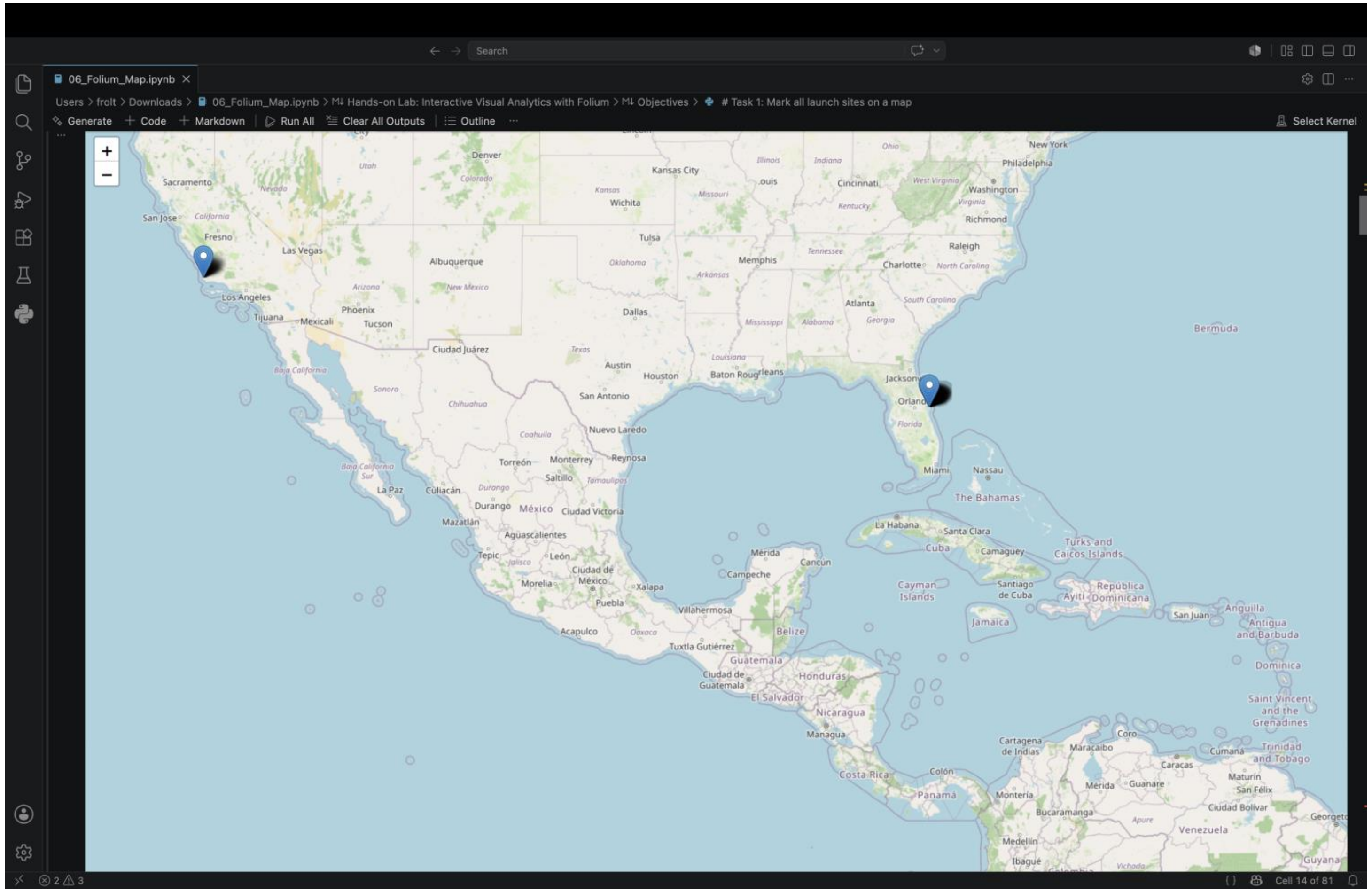
Section 3

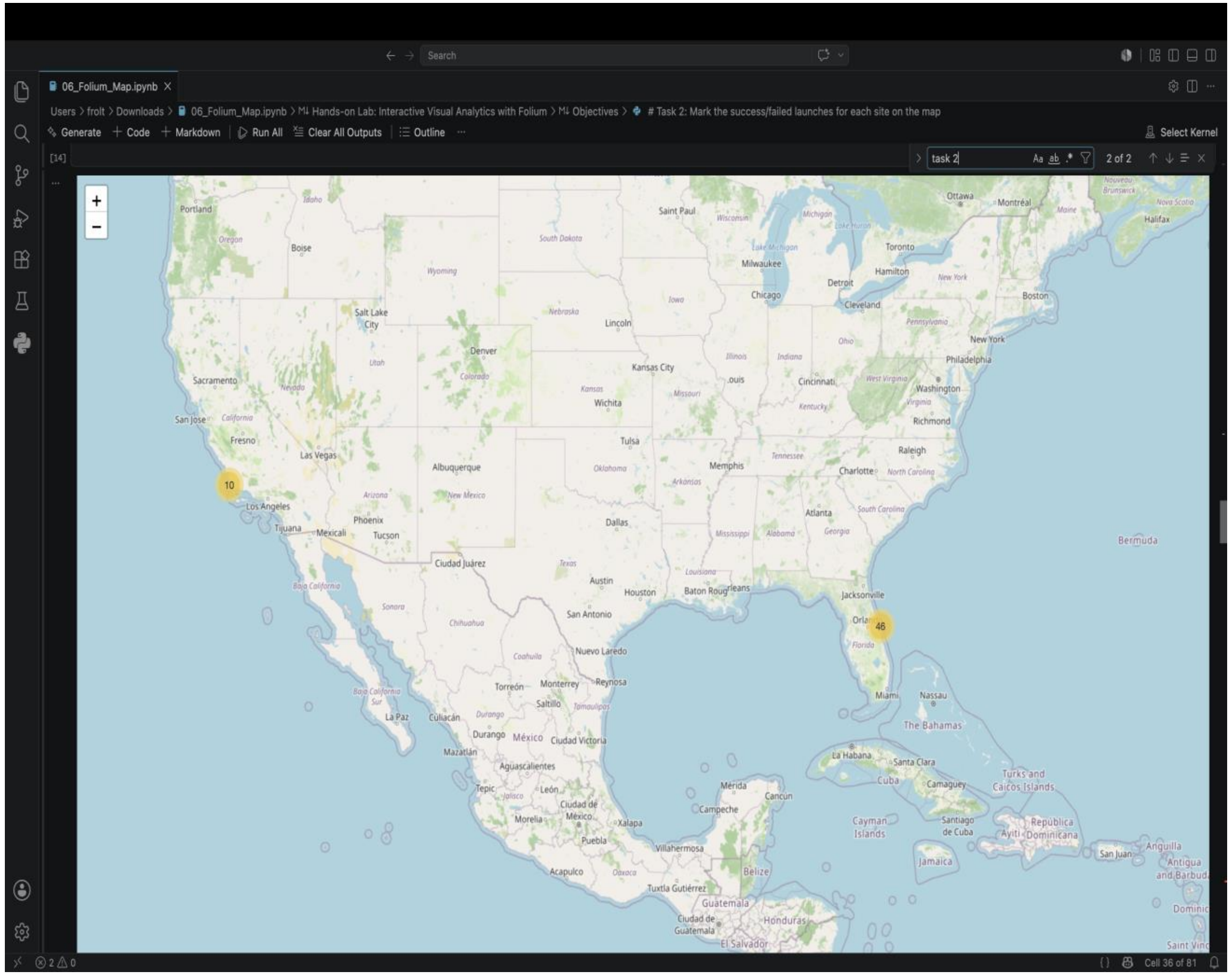
Launch Sites Proximities Analysis

Global Distribution of SpaceX Launch Sites

Explanation -

The Folium map displays the geographic locations of all SpaceX launch sites using circular markers and labels. The interactive map enables users to visualize the spatial distribution of launch facilities, showing that all launch sites are located along the U.S. coastline. Their coastal locations provide safe launch trajectories over the ocean while reducing risks to populated areas, making them well suited for orbital launch operations.





Launch Outcomes by SpaceX Launch Site

Explanation -

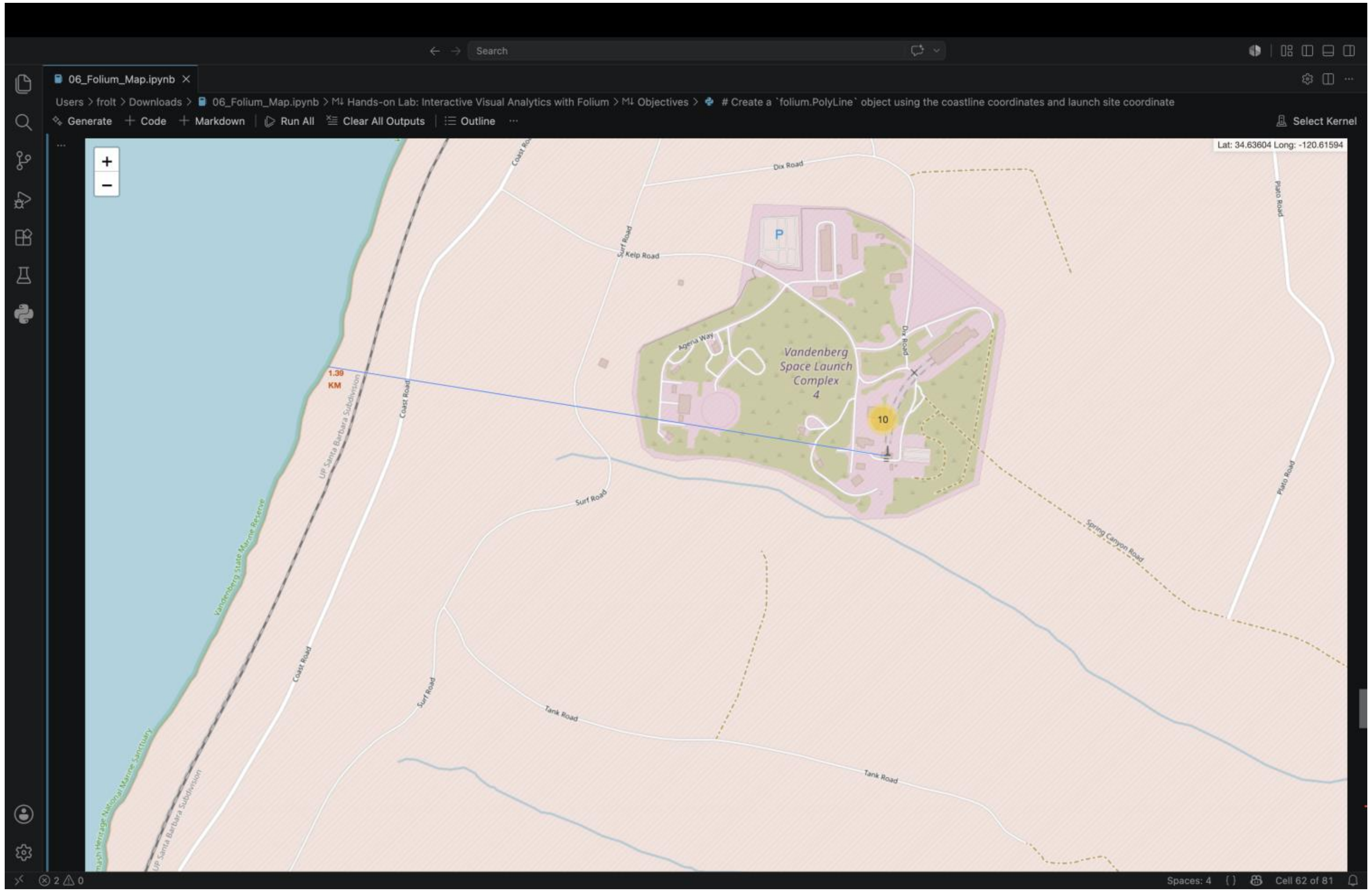
The Folium map displays individual Falcon 9 launch outcomes using color-coded markers. Green markers represent successful launches, while red markers indicate unsuccessful launches. A MarkerCluster groups launches occurring at the same coordinates, making the map easier to explore. The visualization shows that most launch sites contain a higher number of successful launches, reflecting the improvement in Falcon 9 mission performance over time.

Important Elements
Launch site marker identifies the selected SpaceX launch site.
Blue line connects the launch site to the nearest coastline.
Distance label (km) displays the calculated distance.
Interactive map allows users to zoom and inspect the proximity between the launch site and nearby geographic features.

Launch Site Proximity Analysis.- Distance to Coastline

Explanation -

The Folium map highlights the selected launch site and visualizes its proximity to the nearest coastline by drawing a blue distance line. The calculated distance is displayed in kilometers, demonstrating how Folium can be used to measure the distance between a launch site and nearby geographic features. This proximity analysis helps evaluate the surrounding environment and shows that the Vandenberg launch site is located close to the coastline, which is suitable for safe launch operations over the ocean.



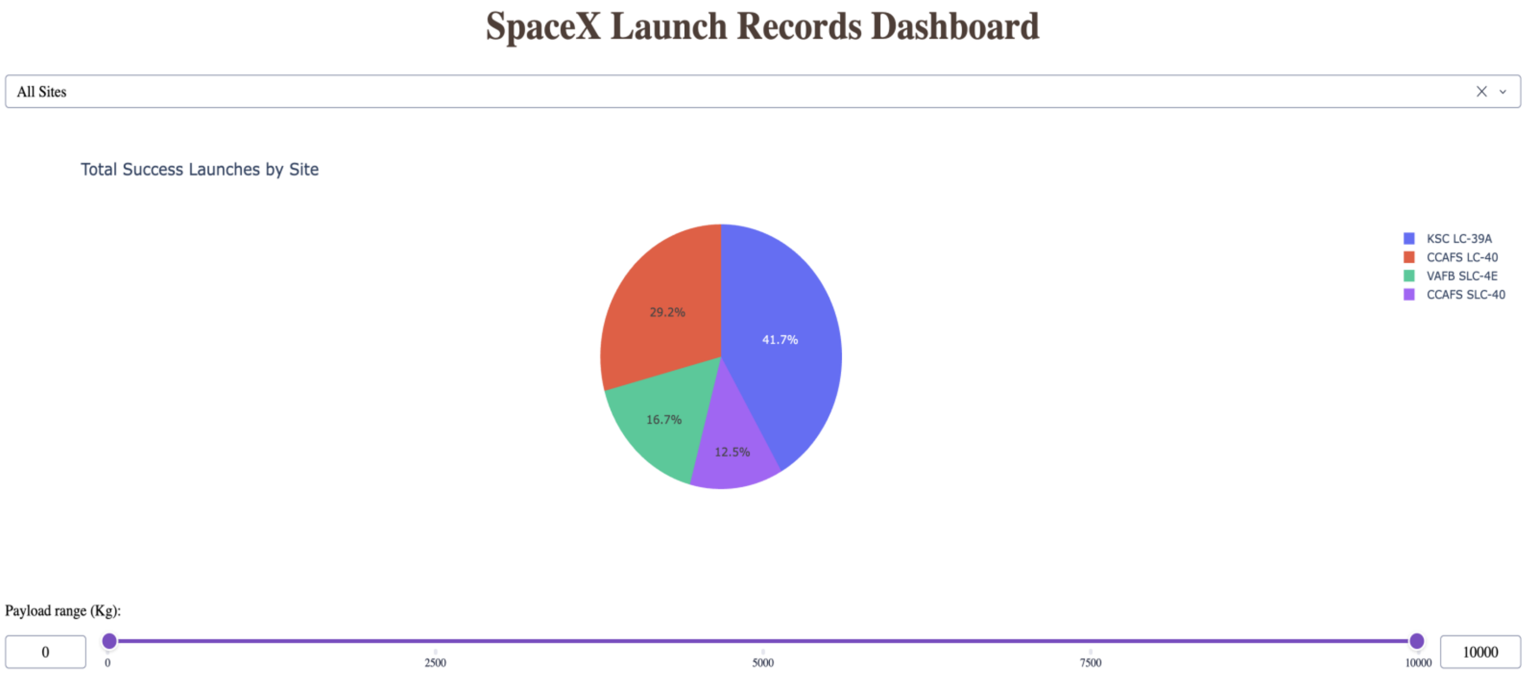
Section 4

Build a Dashboard with Plotly Dash

Total Success Launches by Site

Explanation -

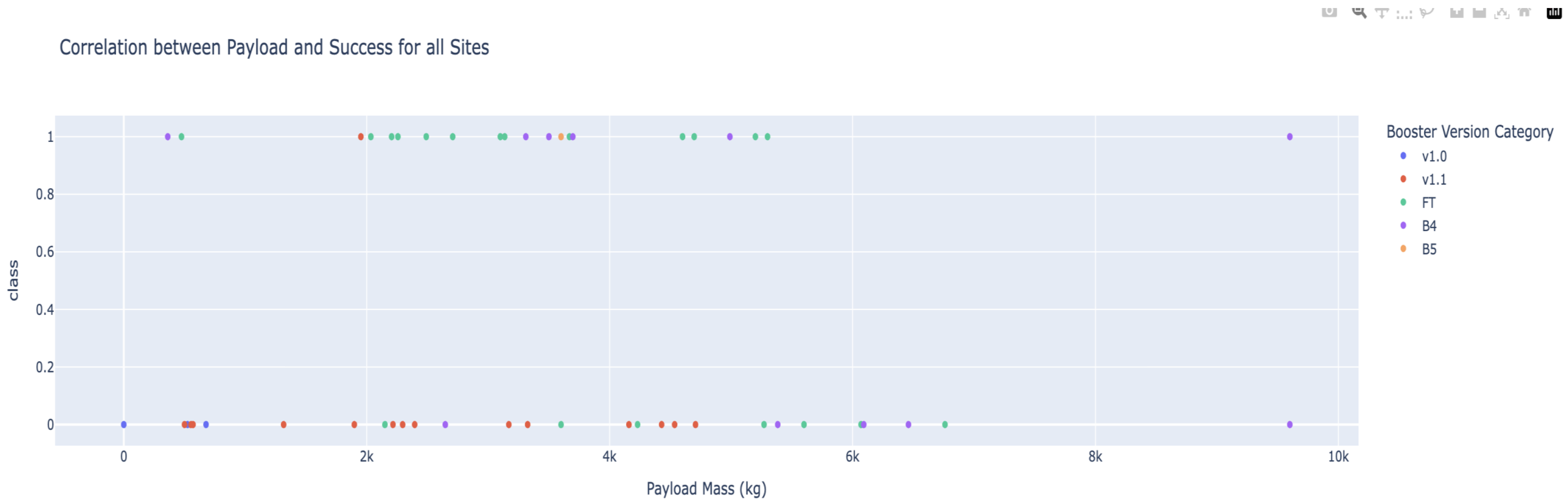
The screenshot shows the pie chart distribution of successful launches across all SpaceX launch sites. KSC LC-39A has the largest share of successful launches (41.7%), followed by CCAFS LC-40 (29.2%), VAFB SLC-4E (16.7%), and CCAFS SLC-40 (12.5%). The findings indicate that KSC LC-39A contributed the highest proportion of successful launches among all launch sites during the analyzed period.



Launch Outcomes for the Selected Launch Site

Explanation -

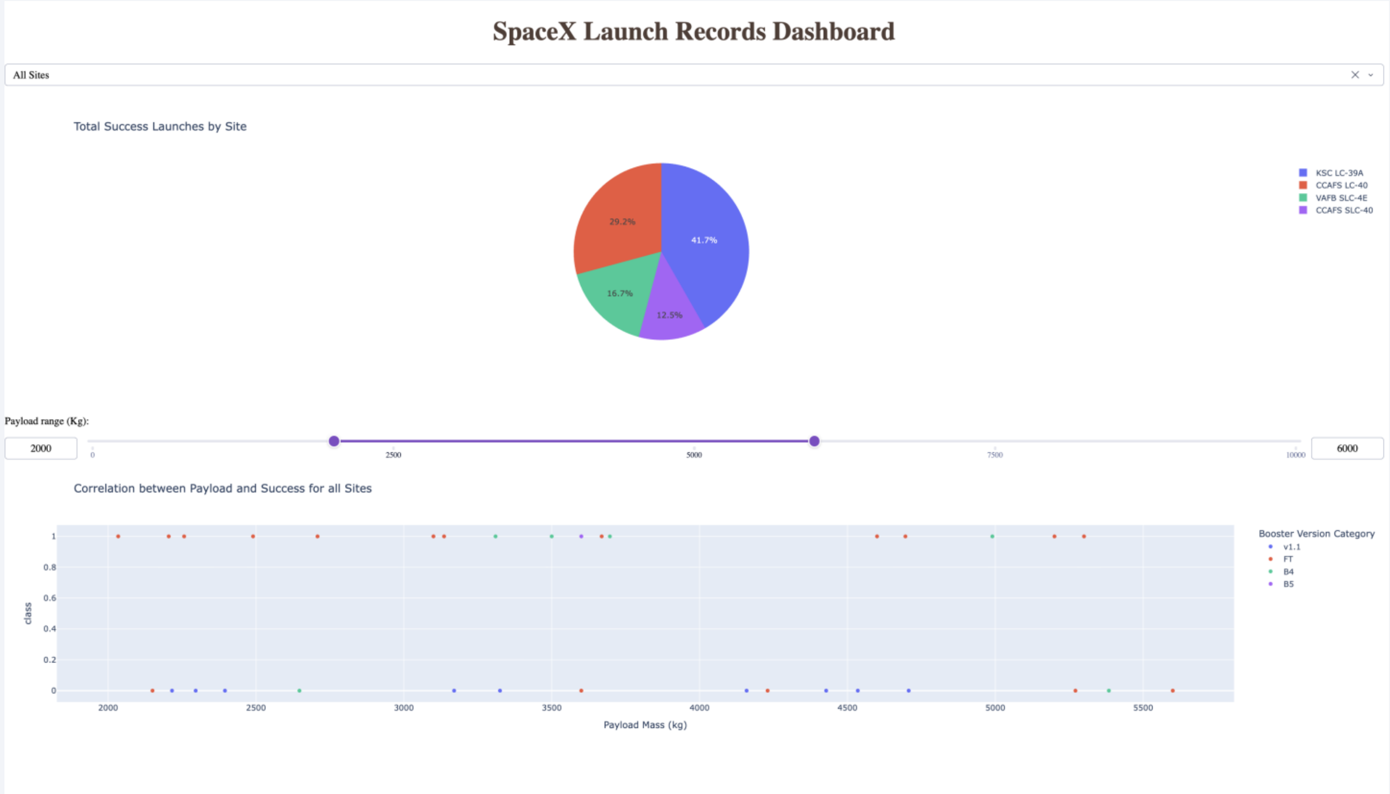
The scatter plot displays the launch outcomes for the selected launch site, showing the relationship between payload mass and landing success. Each point represents a Falcon 9 launch, with the x-axis showing payload mass and the y-axis indicating the landing outcome (0 = unsuccessful, 1 = successful). The visualization shows that successful landings occur across multiple payload ranges, allowing the launch performance of the selected site to be explored interactively.



Correlation Between Payload and Success for All Sites (Payload Range: 2000–6000 kg)

Explanation -

The scatter plot shows the relationship between payload mass and launch success after filtering the payload range to **2,000–6,000 kg**. Each point represents a Falcon 9 launch, where the x-axis indicates payload mass and the y-axis represents the launch outcome (0 = unsuccessful, 1 = successful). Different colors identify the booster version categories. Within the selected payload range, successful launches are more frequent than unsuccessful launches, and FT, B4, and B5 booster versions are associated with many successful missions. The payload range slider enables focused analysis of launch performance for a specific range of payload masses.



Important elements visible in your screenshot

Payload range slider: Filtered to 2,000–6,000 kg.
X-axis: Payload Mass (kg).
Y-axis: Launch Outcome (Class 0 or 1).
Color legend: Booster Version Category (v1.0, v1.1, FT, B4, B5).

Section 5

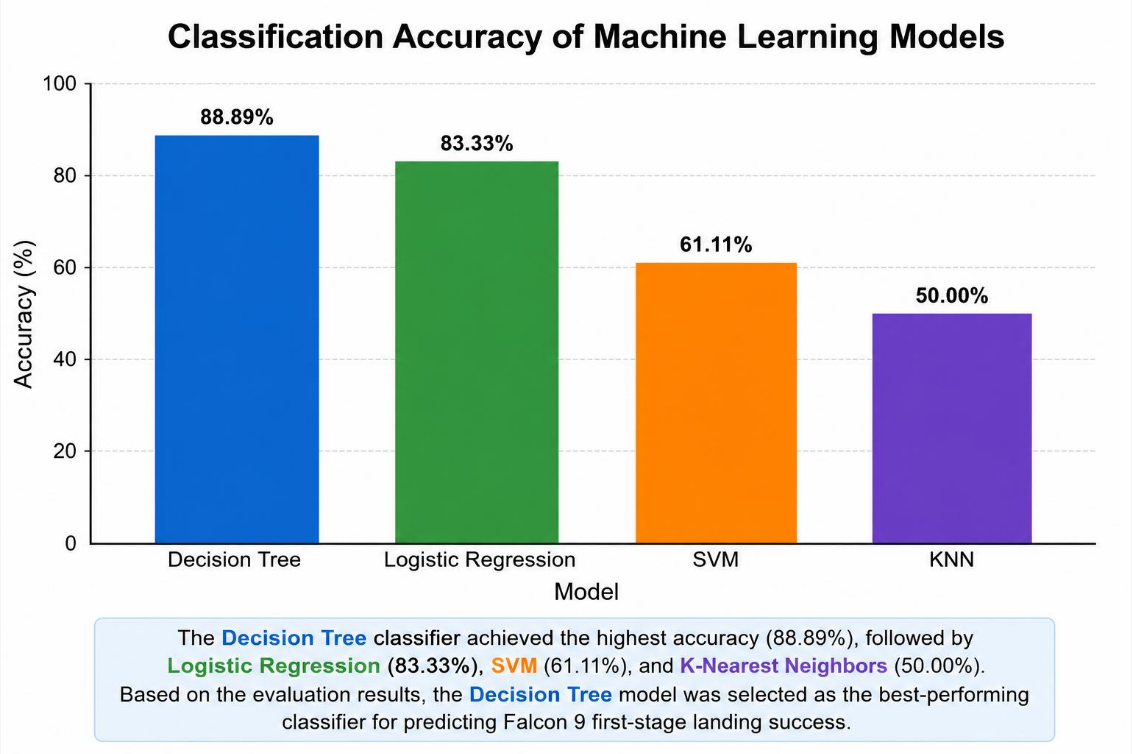
Predictive Analysis (Classification)

Classification Accuracy

Finding

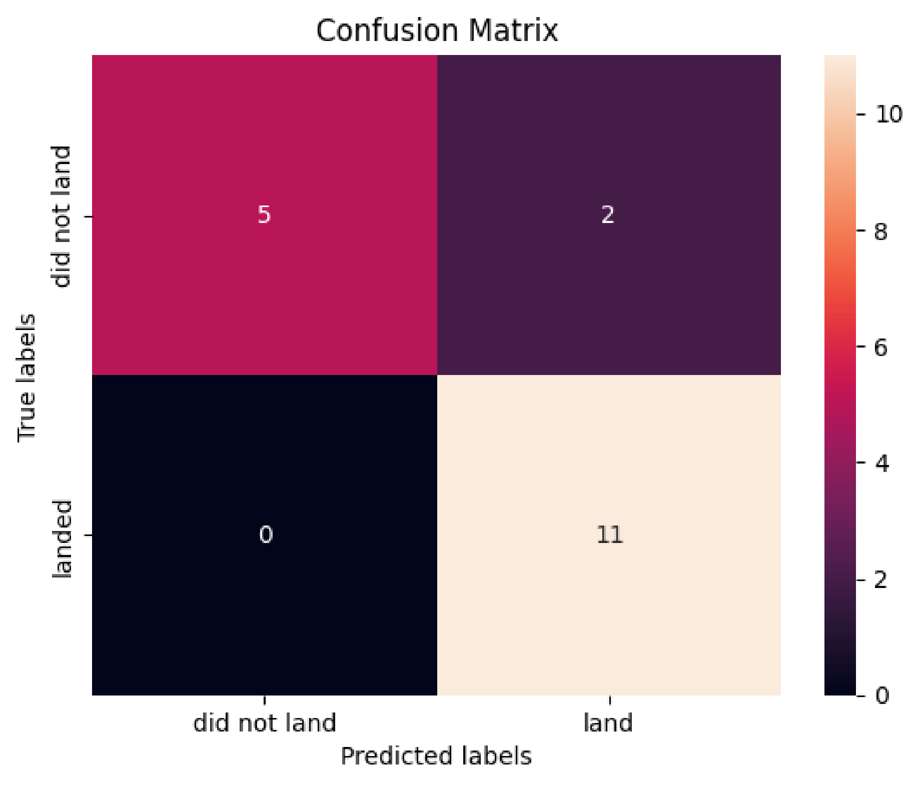
The bar chart compares the classification accuracy of the four machine learning models. The Decision Tree classifier achieved the highest accuracy of 88.89%, followed by Logistic Regression (83.33%), SVM (61.11%), and K-Nearest Neighbors (50.00%). Based on the evaluation results, the Decision Tree model was selected as the best-performing classifier for predicting Falcon 9 first-stage landing success.

Classification Model Accuracy Comparison



Confusion Matrix

Decision Tree Confusion Matrix



Explanation-

The confusion matrix summarizes the performance of the Decision Tree classifier by comparing the predicted and actual landing outcomes. The diagonal cells represent correctly classified launches, while the off-diagonal cells represent incorrect predictions. The high number of correct classifications demonstrates that the Decision Tree model achieved the best overall performance and the highest test accuracy (88.89%) among the evaluated models.

Conclusions

- Exploratory Data Analysis identified launch site, orbit type, payload mass, and booster version as important factors influencing Falcon 9 first-stage landing success.
- Interactive visualizations using Folium maps and the Plotly Dash dashboard enabled effective analysis of launch locations, mission outcomes, and payload-related trends.
- The yearly launch success rate showed a clear upward trend, demonstrating continuous improvements in Falcon 9 landing reliability over time.
- Among the evaluated machine learning models, the Decision Tree classifier achieved the highest classification accuracy (88.89%), making it the best-performing model for predicting Falcon 9 first-stage landing success.

Appendix

PYTHON CODE SNIPPET – EXAMPLE OF THE DECISION TREE MODEL IMPLEMENTATION AND EVALUATION.

SQL QUERY - EXAMPLE QUERY USED TO CALCULATE THE TOTAL PAYLOAD MASS FOR NASA (CRS) MISSIONS.

INTERACTIVE DASHBOARD - PLOTLY DASH APPLICATION FOR ANALYZING LAUNCH SUCCESS BY SITE AND PAYLOAD RANGE.

PROJECT RESOURCES - GITHUB REPOSITORY CONTAINING ALL NOTEBOOKS, SOURCE CODE, AND PRESENTATION ASSETS.

Thank you!